

Ternary mixed-support forcing on the original Erdős–Straus source

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Abstract

Three explicit overlapping observations on one marked divisor source produce a positive signed direction even though all three two-observation restrictions are negative definite. The certificate uses mixed moments only through cubic order and reconstructs an original positive integral witness. Complete source Grams, support predicates, kernels, split weights and prime-power defects are retained. The existing discrete filters extend uniformly to arbitrary growing mixed feature spaces in their actual source metric. A concrete failed ES pair shows why applying those filters after compression would give a false result. This is an unconditional sufficient test and a certified strict example, not a proof of universal Erdős–Straus occupancy.

1 Original arithmetic and the question being tested

Fix an integer $p \equiv 1 \pmod{4}$ and a shell

$$p/4 < a < p/2, \quad R = 4a - p, \quad \gcd(pa, R) = 1, \quad a = \prod_{i=1}^t q_i^{e_i}.$$

Primality is needed for the usual complete-prime-row equivalence, but the individual candidate identities below require only these displayed conditions. An original state α retains (p, a, R, c, β) , the ordered prime list, $-e_i \leq \beta_i \leq e_i$, and

$$u = \prod_i q_i^{e_i + \beta_i}, \quad c \in \{E, M\}, \quad g_E = 4u + 1, \quad g_M = u + a, \quad D_\alpha = \frac{R}{\gcd(R, g_c)}, \quad x_\alpha = D_\alpha^{-1}. \quad (\text{A1})$$

The canonical source has counting measure on these labels, not on distinct residues, split occurrences or sorted triples. In particular,

$$x_\alpha \in \{1, 1/3, 1/5, \dots\}, \quad H = \#\{\alpha : D_\alpha = 1\}. \quad (\text{A2})$$

All failed states have $x \leq 1/3$ because R is positive and odd.

The inherited positive rational triples are

$$E : \left(a, \frac{pa + a^2/u}{R}, \frac{pa + p^2u}{R} \right), \quad M : \left(a, \frac{p(a + a^2/u)}{R}, \frac{p(a + u)}{R} \right). \quad (\text{A3})$$

They obey $(RY - pa)(RZ - pa) = p^2a^2$, hence have reciprocal sum $4/p$. The gcds of the two nonfirst numerators with R are both $\gcd(R, g_c)$: reduce modulo R using $4a \equiv p$ and multiply by the units $a, u, p, 4$ as required. Thus their reduced denominators are exactly D . At a hit let $d_0 = \gcd(a, u)$ and

$$(h, r, s) = (d_0^2/u, u/d_0, a/d_0), \quad a = hrs, \quad u = hr^2, \quad \gcd(r, s) = 1.$$

Prime valuations prove all integrality assertions. The original ordered return is

$$E : \kappa = (pr + s)/R, \quad (a, hsk, phrk), \quad M : \lambda = (r + s)/R, \quad (a, phs\lambda, phr\lambda). \quad (\text{A4})$$

The inverse recovers $u = a^2/(RY - pa)$ in E and $u = pa^2/(RY - pa)$ in M, then every $\beta_i = v_{q_i}(u) - e_i$. These are inherited Type I/II coordinates [1, §2, Propositions 2.2 and 2.6]; no new parametrization is claimed. The middle reversal $u \mapsto a^2/u$ and an exterior ordering bit, when used, remain distinct operations. No positive-Koide restriction is imposed in the new mixed test.

2 One original source, several overlapping observations

Let Ω be a finite set of original states, and set $\mathcal{H} = \mathbb{R}^\Omega$ with its original counting inner product. Take explicit real functions $f_0, \dots, f_{r-1}$ on this same set. Their supports may overlap. Define the synthesis map and its two pullback forms:

$$A : \mathbb{R}^r \rightarrow \mathcal{H}, \quad (Ac)_\alpha = \sum_i c_i f_i(\alpha), \quad K = A^*A, \quad J = A^*(X - \tfrac{1}{3}I)A, \quad Xe_\alpha = x_\alpha e_\alpha. \quad (\text{S1})$$

Thus

$$K_{ij} = \sum_\alpha f_i(\alpha) f_j(\alpha), \quad J_{ij} = \sum_\alpha (x_\alpha - \tfrac{1}{3}) f_i(\alpha) f_j(\alpha). \quad (\text{S2})$$

Every cross term is on the original arithmetic source. Separate labels for two columns do not make their images orthogonal. These are finite signed localizing forms, a classical construction [2, §2, equations (2.3)–(2.4); Theorem 3.2]. The particular support functions and arithmetic certificates below are the new application, not a claim to have invented that construction.

Theorem 1 (Mixed-support positivity and multiplicity). *For the actual forms (S1), $\ker K = \ker A \subseteq \ker J$ and*

$$n_+(J) \leq \text{rank}(\Pi A) \leq H, \quad \Pi e_\alpha = \mathbf{1}_{D_\alpha=1} e_\alpha. \quad (\text{S3})$$

*In particular, a coefficient vector with $c^*Jc > 0$ forces an original hit. If the explicitly proved bound $|(Ac)_\alpha|^2 \leq M(c)$ holds on all possible successful patterns, then*

$$H \geq \left\lceil \frac{3c^*Jc}{2M(c)} \right\rceil \quad (M(c) > 0). \quad (\text{S4})$$

No occupied support, successful state, or unknown positive contribution is assumed in calculating J or in the pattern bound.

Proof. The equality of kernels follows from $c^*Kc = \|Ac\|^2$. On $D = 1$ the multiplier $x - 1/3$ equals $2/3$; on every other state it is nonpositive. Thus

$$c^*Jc \leq \frac{2}{3} \sum_{D_\alpha=1} |(Ac)_\alpha|^2.$$

This proves (S4). On a subspace where J is positive definite, ΠA is injective: a nonzero vector in its kernel would have nonpositive form by the same calculation. Taking dimensions proves (S3). $\square$

Kernels, inverses and actual source norms. The image of A has the quotient norm c^*Kc on $V = \mathbb{R}^r / \ker A$. In this metric $\bar{A} : V \rightarrow \text{im } A$ is an isometry. The inverse on the image is the coefficient class of any solution of $Ac = f$; all solutions are $c_0 + \ker A$. Exact row reduction gives a solution and every kernel generator. If K is invertible, the inverse is $K^{-1}A^*$. For arbitrary original f the discarded component is explicitly $(I - P_A)f$, where P_A is the orthogonal projection onto $\text{im } A$. The full return keeps this component, rather than identifying all original states with an r -coordinate observation.

If several independently supplied support functions are first assembled by addition, the larger synthesis map $\bigoplus_i \mathbb{R}^{\text{supp } f_i} \rightarrow \mathbb{R}^\Omega$ has one fibre-sum condition at each original coordinate. Its kernel is the full space of those sums zero, with primitives $\sum_{i \neq i_0} v_i(e_i - e_{i_0})$ at that coordinate. Cancellation of values retains the support union in the SplitZero carrier; it is not external absence. Restricting this larger map to the specified column functions gives (S1).

The source comparison actually used. The pinned Zeta proof calculates a full joint covariance $\begin{pmatrix} K & C \\ C^* & B \end{pmatrix}$, its Schur complements, and the original relation-valued minimum-section correction [3, JSA12–JSA20]. Our positive K and signed J are not those analytic Gamma matrices. We apply their same-source/cross-term discipline, and prove the signed elimination needed here directly. No Gamma asymptotic, theta-source norm, or analytic endpoint claim is imported by analogy.

3 The exact third-support contribution

The following calculation keeps the first column and both other support columns in their stated order. Write

$$J = \begin{pmatrix} -a & b^* \\ b & C \end{pmatrix}, \quad a > 0.$$

Then the reversible coordinate change

$$(t, z) \mapsto (s, z), \quad s = t - b^*z/a, \quad (t, z) = (s + b^*z/a, z) \quad (\text{S5})$$

gives

$$(t, z)^* J(t, z) = -a|s|^2 + z^* S z, \quad S = C + b b^*/a. \quad (\text{S6})$$

This is a congruence with the original signed form, not a positive-norm orthogonal projection. It preserves inertia and has no unspecified scale.

Theorem 2 (Strict ternary criterion). *For a real symmetric 3×3 form whose three diagonal entries are negative and whose three two-coordinate principal determinants are positive,*

$$\boxed{J \text{ has a positive direction} \iff \det J > 0.} \quad (\text{S7})$$

When this holds its inertia is $(1, 2, 0)$: exactly one positive and two negative directions. All two-column restrictions remain negative definite.

Proof. The two diagonal entries of S in (S6) are negative by the principal-minor conditions involving the first coordinate. Also $\det J = -a \det S$. A 2×2 symmetric matrix with negative diagonal entries is negative semidefinite exactly when its determinant is nonnegative. A negative determinant gives one positive and one negative direction. Equation (S6) proves all assertions. Zero pivots and vanishing principal determinants are excluded from this strict statement, not inverted; (S3)–(S4) still apply there using their actual kernels. $\square$

The mixed condition is $S_{12}^2 > S_{11}S_{22}$, with the complete conditional entry $S_{12} = C_{12} + b_1b_2/a$. Dropping C_{12} or replacing the three-column space by the common-polynomial diagonal restriction changes the problem. The strict comparison refers to the three given coordinate subspaces. It does not assert that every imaginable two-function reparametrization fails: one may name an already combined function as a new column, provided its full three-term decomposition remains attached.

4 A genuine three-support certificate on the original hard shell

The arithmetic integer and its exact factorization are

$$\begin{aligned} p &= 2671180768668258904496300170662649, \\ a &= 667795192167064726124075042665696 = 2^5 431^3 457^3 7229^4, \\ R &= 135, \quad p = 4a - 135 \equiv 289 \pmod{840}. \end{aligned} \quad (\text{E1})$$

The attached complete-order certificate proves p prime; the arithmetic positivity proof below already holds without appealing to its primality. This is the second hard shell from the preceding tranche [4], reused for a genuinely different support test rather than called a new prime. There are $2 \cdot 11 \cdot 7 \cdot 7 \cdot 9 = 9702$ original states.

Define the following supports using only the channel and original exponent:

$$\mathcal{E} = \{c = E\}, \quad \mathcal{B} = \{\beta_{457} > 0\}, \quad f_0 = 1, \quad f_1 = x\mathbf{1}_{\mathcal{E}}, \quad f_2 = x\mathbf{1}_{\mathcal{B}}. \quad (\text{E2})$$

These supports are the whole source, the exterior channel, and the upper half of one actual exponent interval. The latter overlaps both channels. It means $v_{457}(u) > 3$, not membership in a set of already known witnesses.

For $i, j \in \{0, 1\}$, let S_k^{ij} sum $\gcd(R, g)^k$ over the cell $(\mathbf{1}_{\mathcal{E}}, \mathbf{1}_{\mathcal{B}}) = (i, j)$. All data through cubic order are as follows:

(i, j)	S_0	S_1	S_2	S_3
(0, 0)	2772	19480	435596	14842144
(0, 1)	2079	14819	332487	11329307
(1, 0)	2772	19474	435524	14841442
(1, 1)	2079	14805	368151	17899341

(E3)

These are common-state mixed moments, not products of marginal moments. Writing $S_k^E = S_k^{10} + S_k^{11}$ and similarly for B , the exact entries are

$$\begin{aligned} (3R^3J)_{00} &= R^2(3S_1 - RS_0), & (3R^3J)_{01} &= R(3S_2^E - RS_1^E), \\ (3R^3J)_{02} &= R(3S_2^B - RS_1^B), & (3R^3J)_{11} &= 3S_3^E - RS_2^E, \\ (3R^3J)_{22} &= 3S_3^B - RS_2^B, & (3R^3J)_{12} &= 3S_3^{11} - RS_2^{11}. \end{aligned} \quad (\text{E4})$$

The complete matrix is

$$3R^3J = \begin{pmatrix} -20121056100 & -299246400 & -256139010 \\ -299246400 & -10273776 & 3997638 \\ -256139010 & 3997638 & -6900186 \end{pmatrix}. \quad (\text{E5})$$

Its three two-coordinate principal determinants are respectively

$$117170815341873600, \quad 73231837162654500, \quad 54909855743292,$$

and its full determinant is

$$\boxed{799917306693954510200400 > 0.} \quad (\text{E6})$$

Thus every two-component restriction fails the signed positivity test, whereas all three components have a positive direction. The usual scalar degree-one test on $(1, x)$ fails on the full source, on E alone and on M alone; their independently recalculated exact forms are in the certificate. This is stronger than a counterexample only to one aggregated channel test.

A small rational trial vector suffices; no eigenvector approximation enters:

$$F_\alpha = \frac{-1 + 36x_\alpha \mathbf{1}_E + 36x_\alpha \mathbf{1}_B}{71}. \quad (\text{E7})$$

At a hit, its four possible values are $-1/71, 35/71, 35/71, 1$. Hence their squares are at most one, independently of which patterns hit. Using (E5) yields

$$\boxed{\frac{3}{2} \sum_\alpha (x_\alpha - \frac{1}{3}) F_\alpha^2 = \frac{607462}{1890375} > 0.} \quad (\text{E8})$$

This proves that an original integral witness exists. If just the 12 cross entry is deleted, the same calculation incorrectly becomes $-1639754/17013375$. The retained cross term changes the conclusion.

The ordinary source metric, separately, is

$$K = \frac{1}{R^2} \begin{pmatrix} 176818950 & 4627665 & 3999240 \\ 4627665 & 803675 & 368151 \\ 3999240 & 368151 & 700638 \end{pmatrix}. \quad (\text{E9})$$

It is positive definite: source rows $(M, u = 1)$, $(E, u = 1)$, and $(M, u = 457^4)$ give independent evaluations of the three columns. For $F \in \text{im } A$, recover $c_0 = F(M, 1)$, $c_1 = (F(E, 1) - c_0)/x(E, 1)$, and $c_2 = (F(M, 457^4) - c_0)/x(M, 457^4)$. The full membership test is then $F_\alpha = c_0 + c_1 f_1(\alpha) + c_2 f_2(\alpha)$ for every original state. This gives an explicit image inverse as well as the metric inverse $K^{-1}A^*$.

Finite census, primality, and actual positive return. The retained prime exponents are $(5, 3, 3, 4)$. Direct divisor enumeration and a separate bounded multiplication in $\mathbb{Z}[(\mathbb{Z}/135\mathbb{Z})^\times]$, marked by whether the 457 exponent exceeds three, give exactly the same table (E3). The two channel distributions in defect order 1, 3, 5, 9, 15, 27, 45, 135 are

$$E : (3, 202, 266, 403, 336, 617, 1212, 1812), \quad M : (0, 210, 270, 406, 328, 616, 1208, 1813).$$

All three hits have

$$\beta = (5, j, 3, -4), \quad j \in \{-2, 0, 2\}, \quad u = 2^{10} 431^{j+3} 457^6.$$

For the first, $j = -2$, the recovered data include

$$u = 4020447794899891321856, \quad h = 431, \quad r = 3054207776, \quad s = 507303010196865101041,$$

$$\kappa = 60432156109394470448106286277616196397479, \quad (X, Y, Z) = (a, hsk, phrk). \quad (\text{E10})$$

The identities $a = hrs$, $u = hr^2$, $135\kappa = pr + s$ and (A4) prove positivity and the reciprocal equation. Exact expanded denominators and all inverse checks are serialized, not rounded.

The supplied primality proof is independently executed: all eight nontrivial nodes give the complete factorization of $n - 1$ and a base b satisfying $b^{n-1} = 1 \pmod{n}$ and $\gcd(b^{(n-1)/q} - 1, n) = 1$ for every prime $q \mid n - 1$. For each prime divisor $l \mid n$ these conditions force the order modulo l to be $n - 1$, hence $l \geq n$ and n is prime. Fifteen terminal primes are checked by exhaustive trial division; there are 23 total nodes. The large prime and its primality certificate are inherited from [4]; only the new mixed form is being added.

5 Finite certificates lifted back to all split supports

For an actual budget split $e = f + b$, let $q(\xi, \eta, c) = (\xi + \eta, c)$. Its complete inverse fibre is

$$\max(-f_i, \beta_i - b_i) \leq \xi_i \leq \min(f_i, \beta_i + b_i), \quad \eta = \beta - \xi,$$

with size the product n_α of these interval lengths. Equip the occurrence space with weight $1/n_{q\omega}$ and define

$$(Uv)_\omega = v_{q\omega}, \quad (EF)_\alpha = \frac{1}{n_\alpha} \sum_{q\omega=\alpha} F_\omega.$$

Grouping each finite fibre proves $U^* = E$, $EU = I$, and $\|F\|^2 = \|EF\|^2 + \|F - UEF\|^2$. The kernel of E consists of fibre sums zero, with the explicit star-difference primitives at every fibre. The full source image retains those kernel coordinates on return. For any original multiplier $h(X)$ and actual feature synthesis A ,

$$\boxed{(UA)^* h(\tilde{X})(UA) = A^* h(X) A.} \quad (\text{W1})$$

Here $\tilde{X}$ repeats the original coefficient and every support predicate is pulled back through q . In particular $\beta_{457} > 0$ is tested on $\xi + \eta$, not on either summand alone.

The hard-shell check uses $f = (5, 2, 2, 4)$ and $b = (0, 1, 1, 0)$. There are 44550 occurrences over 9702 original states. Seven raw hit occurrences have weighted mass three. Both K and J return exactly to (E5),(E9). Replacing $\beta_{457} > 0$ by $\xi_{457} > 0$ gives a different form, recorded as a rejected false substitution. Prime-power boundary degrees also pull back unchanged. In the earlier module $J_\alpha = \mathbb{Q}[z_\ell]/(z_\ell^{v_\ell(D_\alpha)+1})_\ell$, top insertion s and constant extraction ϵ still obey $\epsilon s = \Pi$; hence the mixed arithmetic comparison is $\epsilon s A = \Pi A$. All supported top lines at failures remain in the source even though this particular receiving image is zero. The new quadratic form is not relabelled as the dimension of the older residue-cohomology quotient.

6 Uniform mixed control and the order of compression

The previous discrete arithmetic filters are

$$\Phi_j(x) = x \prod_{\nu=1}^j \frac{(2\nu+1)x-1}{2\nu}, \quad \varepsilon_j = \frac{1}{2^j j! (2j+3)}.$$

At $x = 1/(2k+1)$, the value is zero for $1 \leq k \leq j$, otherwise $(-1)^j \binom{k-1}{j}/(2k+1)^{j+1}$; it is one at $x = 1$. This proves the inherited bounds $\Phi_{2j+1} \leq \mathbf{1}_{\{1\}} \leq \Phi_{2j}$ and uniform error ε_j directly [4]. For every actual mixed synthesis A ,

$$\begin{aligned} A^* \Phi_{2j+1}(X) A &\preceq A^* \Pi A \preceq A^* \Phi_{2j}(X) A, \\ 0 &\preceq (-1)^j (A^* \Phi_j(X) A - A^* \Pi A) \preceq \varepsilon_j K. \end{aligned} \quad (\text{U1})$$

These are inequalities of full forms, including off-diagonal entries, not entrywise inequalities. To prove them, evaluate both sides on any coefficient vector and sum the pointwise filter inequality weighted by $|(Ac)_\alpha|^2$.

The conclusion is independent of the number and sizes of overlapping supports. For any countable prime-indexed family, use the actual metric quotient $V_p = \mathbb{C}^{r_p} / \ker A_p$ with Gram K_p . The induced maps A_p are isometries; their direct sum is an isometry into the original counting Hilbert space. Consequently on $\bigoplus_p V_p$,

$$\left\| \bigoplus_p \bar{A}_p^* (\Phi_j(X_p) - \Pi_p) \bar{A}_p \right\| \leq \varepsilon_j. \quad (\text{U2})$$

No uniform bound on r_p , support multiplicity, or the original row mass is needed. The metric quotient and its inverse on the image are retained; this is not a rescaling of the original successful-state count. If the A_p have proper images, $A_p^* \Pi_p A_p$ is a compression of the success projection, not the projection itself and not its original trace. The forgotten orthogonal sector remains specified in (S1).

Theorem 3 (The order-of-composition correction). *Let $\bar{A}$ be an isometry onto a feature subspace and $P_A = \bar{A}\bar{A}^*$. For $C = \bar{A}^* X \bar{A}$,*

$$\boxed{\bar{A}^* X^2 \bar{A} - C^2 = \bar{A}^* X (I - P_A) X \bar{A} = R_A^* R_A, \quad R_A = (I - P_A) X \bar{A}.} \quad (\text{U3})$$

It vanishes exactly when the feature subspace is X -invariant. In general one cannot replace the matrices in (U1) by $\Phi_j(C)$.

Proof. Insert $I = P_A + (I - P_A)$ between the two copies of X . The first summand is C^2 , the other is the displayed squared norm. Vanishing is equivalent to $(I - P_A) X \bar{A} = 0$, which is precisely invariance. Self-adjointness then also makes the orthogonal complement invariant. $\square$

The relation R_A is retained with its source norm, as in the source's minimum-section corrections; it is not discarded merely because its projection to the feature quotient is zero. For the hard example the verifier calculates the full nonzero positive semidefinite coordinate defect $M_2 - M_1 K^{-1} M_1$, where $M_i = A^* X^i A$.

The danger has an actual ES countercheck. In the empty shell $p = 37, a = 18, R = 35$, the exterior states $u = 1, 12$ give

$$Q_1 = (18, 198/7, 407/7), \quad Q_2 = (18, 99/5, 2442/5),$$

with $D_1 = 7, D_2 = 5$. Compress their counting source onto its constant line. Then $C = 6/35$, so

$$\boxed{\bar{A}^* \Phi_3(X) \bar{A} = 0, \quad \Phi_3(C) = 17/343000 > 0.} \quad (\text{U4})$$

The second expression would falsely report positivity. The quadratic relation norm in (U3) is $1/1225$. Applying the filter first and then taking the mixed Gram is therefore essential, even though each operation separately is defined.

7 Evidence, scope and next arithmetic bound

The default standard-library verification constructs all 9702 hard-shell states and checks their rational identities, return denominators and ordered inverses. It independently reconstitutes the four mixed god-moment rows by bounded group multiplication, checks every two-coordinate negative form and the full positive direction, and records all three integral witnesses. It tests 729 symmetric integer forms including singular boundaries, 256 nonorthogonal support dictionaries at five filter degrees, the actual split with 44550 occurrences, and six rejected false transformations.

The additional scan covers all 9531 first-half shells at the 54 primes $p \equiv 1 \pmod{12}$, $p \leq 1500$: 412838 canonical states and 20177 choices of the support $\beta_q > 0$. There are 1693 certified choices covering the 685 occupied shells. There are no strictly ternary-only choices in this small range; the fixed large example supplies the strict improvement. These counts are not a new prime-coverage result, and the large example is not claimed least. All inputs for the printed positive certificate are exact integers or rationals; floating-point exploration is not used in its verification.

The new full three-support form, its explicit original arithmetic return, and the uniform mixed-form transport are the contributions of this tranche. Signed localizing forms, Schur congruences and compression identities have classical antecedents. The scoped literature comparison did not establish historical priority for the precise ES example; none is asserted.

For every hypothetical false prime, all such original mixed forms must be negative semidefinite, including their ternary determinants and higher mixed minors. The unresolved universal step is to force a positive direction in at least one shell by an arithmetic estimate on the actual mixed moments. Pairwise estimates omit a demonstrably relevant condition, but the correct full condition is not yet proved positive for every prime. Uniform convergence (U2) controls all retained observations without supplying that arithmetic lower bound. No ES, RH, or global geometric conclusion is claimed.

References

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- [4] The Clankers (2026). *Discrete-spectrum positivity and hard-shell continuations*. The supplied package `es_discrete_positivity_bundle.zip`, `core.tex`, sections on exact one-sided filters, hard-shell examples, and support refinements; `certificates/second_hard_prime.json`. Source arithmetic, large prime and order certificate retained with attribution. The present verifier recalculates the new mixed application independently.